using their absorption spectral data. The absorption wave numbers of complexes $[TiBr_6]^{3-}$, $[TiF_6]^{3-}$ and $[Ti(H_2O)_6]^{3+}$ are 12500, 19000 and 20000 cm^{-1} respectively. The energy associated with the absorbed wave numbers of the light, corresponds to the crystal field splitting energy (Δ) and is given by the following expression,

$$\Delta = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}$$

where h is the Plank' s constant; c is velocity of light, $\bar{\upsilon}$ is the wave number of absorption maximum which is equal to 1/λ

$[TiBr_6]^{3-}$	$[TiF_6]^{3-}$	$[Ti(H_2O)_6]^{3+}$
$\Delta = hc\bar{\nu}$	$\Delta = hc\bar{\nu}$	$\Delta = hc\bar{\nu}$
$= (6.626 \times 10^{-34}\ Js) \times (3 \times 10^{8}\ ms^{-1}) \times (12500 \times 10^{2}\ m^{-1})$	$= (6.626 \times 10^{-34}\ Js) \times (3 \times 10^{8}\ ms^{-1}) \times (19000 \times 10^{2}\ m^{-1})$	$= (6.626 \times 10^{-34}\ Js) \times (3 \times 10^{8}\ ms^{-1}) \times (20000 \times 10^{2}\ m^{-1})$
$= 248475 \times 10^{-24}\ J$	$= 377682 \times 10^{-24}\ J$	$= 397560 \times 10^{-24}\ J$
$= 2.48 \times 10^{-22}\ kJ$	$= 3.78 \times 10^{-22}\ kJ$	$= 3.98 \times 10^{-22}\ kJ$
To express Δ on a per mole basis, multiply it by Avogadro number	To express Δ on a per mole basis, multiply it by Avogadro number	To express Δ on a per mole basis, multiply it by Avogadro number
$= (2.48 \times 10^{-22}\ kJ) \times (6.023 \times 10^{23} mol^{-1})$	$= (3.78 \times 10^{-22}\ kJ) \times (6.023 \times 10^{23} mol^{-1})$	$= (3.98 \times 10^{-22}\ kJ) \times (6.023 \times 10^{23} mol^{-1})$
$= 149.4\ kJ\ mol^{-1}$	$= 227.7\ kJ\ mol^{-1}$	$= 239.7\ kJ\ mol^{-1}$

From the above calculations, it is clear that the crystal filed splitting energy of the Ti^{3+} in complexes,the three ligands is in the order; $Br^- < F^- < H_2O$. Similarly, it has been found form the spectral data that the crystal field splitting power of various ligands for a given metal ion, are in the following order

$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- \approx urea < ox^{2-} < H_2O < NCS^- < EDTA^{4-} < NH_3 < en < NO_2^- < CN^- < CO$

The above series is known as spectrochemical series. The ligands present on the right side of the series such as carbonyl causes relatively larger crystal field splitting and are called strong ligands or strong field ligands, while the ligands on the left side are called weak field ligands and causes relatively smaller crystal field splitting.

XII U5 Coordination jagan.indd 156 22-12-2022 19:51:05